No weight extracts $\sum_{n<N} \Lambda(n)\mu(N-n)$: a no-go for the divisor-switch route, with the exact identities of its design space

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Abstract

Let N be a large even integer and $C(N) = \sum_{n<N} \Lambda(n)\mu(N-n)$. The Huang–Li reduction of binary Goldbach to Elliott–Halberstam [3] consumes its hypothesis through a Möbius-weighted correlation sum in the fixed class $n \equiv N\ (k)$, carrying a weight w_k; the two weights that occur are $w_k = 1$, for which the sum is unconditionally small, and $w_k = \log k$, for which it is equivalent to Goldbach itself [6]. It is natural to ask whether some weight sits between them: one whose complete divisor sum is still cheap but whose main-term coefficient is still of size 1, and which would therefore extract $C(N)$ from Bombieri–Vinogradov alone.

We show that no such weight exists. Writing $b = \mu * w$, the two requirements pull in opposite directions — extraction needs mass in b at the truncation point $K = N^{\theta'}$ (Lemma 15), while the residual is accessible to Bombieri–Vinogradov only for b supported below $N^{\theta'-1/2}$ (Lemma 17) — and the two thresholds are separated by a factor $N^{1/2}$. Theorem 18 converts this into the bound $\|b\|_1/|B_w| \gg \exp(c_1\sqrt{(1/2)\log N})$, which exceeds every power of $\log N$, and which destroys the extraction *even when the hard input is granted for free* — the switch identity carries a third term T_w, of size N, and the theorem hands over the bound on it that only EH_μ could supply; Theorem 21 shows the separation survives granting $EH(N^{\theta_E})$ for any fixed $\theta_E < 1$, so the obstruction is not a shortage of level. The most natural family of weights, $w_k = f(\log k)$ with f polynomial, falls outside the support hypothesis and is closed separately (Proposition 23): its complete part is a nonnegative Chen-type sum, never $o(N)$, and the tuning that would cancel it is circular. For completeness we record the exact position of the other classical mechanism: the circle method applied directly to $C(N)$ has zero margin (Proposition 26), both standard pairings lying at or above the trivial bound, with Parseval supplying the floor.

The design space is first laid out exactly. Seven finite rearrangements (Propositions 3–12) identify the per-modulus discrepancy as a dilate of a Möbius–prime correlation, show that the signs $\mu(k)$ cancel out of the $\log k$ branch exactly, and exhibit the Goldbach count as a signed sum of nonnegative layers. These identities are what make the no-go a statement about a space rather than about two examples.

Net progress toward Goldbach: zero. The value of a no-go of this kind is that it closes a design space rather than leaving it to be re-explored.

1 Introduction

1.1 The setting

Let N be a large even integer, $\theta' \in (1/2, 1)$ fixed, and $K = N^{\theta'}$. Put

$$C(N) = \sum_{n<N} \Lambda(n)\mu(N-n), \qquad A(N;k) = \sum_{\substack{n<N \\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n), \tag{1}$$

and for a weight $w : \mathbb{N} \to \mathbb{R}$,

$$T_w \;=\; \sum_{\substack{k<K\\(k,N)=1}} \mu(k)\, w_k \Big[A(N;k) - \frac{C(N)}{\varphi(k)} \Big]. \tag{2}$$

The functional E_3 of [3] is T_w at $w_k = \log k$; E_4 is not T_w at $w_k = 1$ but is obtained from it by partial summation, an inner factor $\log(N-n)$ separating the two [6, Cor. 2]. E_4 is unconditionally $\ll_A N(\log N)^{-A}$, and E_3 is, by an unconditional identity, equivalent to Huang–Li's equation (22) and hence already gives binary Goldbach [6]. This paper closes the space between the two weights.

Throughout, φ is Euler's function, $\mathrm{rad}(u) = \prod_{p|u} p$, $c, c_1, \ldots$ are positive absolute constants, and

$$\mathfrak{A}(N) = \prod_{q \nmid N} \Big(1 - \frac{1}{q(q-1)}\Big), \qquad \mathfrak{S}(N) = 2 \prod_{p>2} \Big(1 - \tfrac{1}{(p-1)^2}\Big) \prod_{\substack{p|N\\p>2}} \Big(1 + \tfrac{1}{p-2}\Big). \tag{3}$$

1.2 The design space

Let w be arbitrary and let $b := \mu * w$, so that $w_k = \sum_{d|k} b_d$; every weight has this form, and b is determined by w. Put

$$B_w := \sum_{\substack{k<K\\(k,N)=1}} \frac{\mu(k) w_k}{\varphi(k)}, \qquad \|b\|_1 := \sum_{u<N} |b_u|. \tag{4}$$

Running the divisor switch of [6] with the weight w gives the identity

$$B_w \cdot C(N) \;=\; \underbrace{\sum_{u<N} \Lambda(N-u)\mu^2(u)\, b_u}_{\text{complete part}} \;-\; \underbrace{\mathcal{R}_w}_{\text{residual}} \;-\; \underbrace{T_w}_{\text{the object itself}} \;+\; O\Big(N^{o(1)}\Big), \tag{5}$$

the complete part being evaluated by Lemma 2 below, which identifies b as exactly the complete divisor transform.

Note 1 (the third term is not a remainder). The term T_w in (5) is (2) itself, and it is of size N: at $w = \log$ it *is* $E_3(\alpha)$, which [6, Thm. 3] evaluates as $\tilde{r}(N) - \mathfrak{S}(N)(N - C(N)) + O_A(N(\log N)^{-A})$. It cannot be absorbed into the $O(N^{o(1)})$, and writing (5) without it turns a rearrangement into a false statement — the two sides then differ by $0.44N$, $0.38N$, $0.32N$ at $N = 2\cdot 10^5$, $4\cdot 10^5$, $8\cdot 10^5$ with $\theta' = 0.56$, which are exactly the measured values of $|T_{\log}|/N$ at those N (Measurement 7). With T_w restored the identity holds to 10^{-16} relative.

Keeping it explicit is what makes Theorem 18 say something. The identity by itself yields nothing about $C(N)$: an unconditional bound on T_w is exactly what the whole problem asks for, and the only weight for which one is known is $w = 1$, by [6, Thm. 1]. The theorem below therefore *grants* the bound on T_w and shows the extraction fails anyway.

Lemma 2. *For squarefree u, $\sum_{k|u} \mu(k) w_k = \mu(u)\, b_u$.*

Proof. $\sum_{k|u} \mu(k) \sum_{d|k} b_d = \sum_{d|u} b_d\, \mu(d) \sum_{j|(u/d)} \mu(j) = \sum_{d|u} b_d\, \mu(d)\, [\, u/d = 1\,] = \mu(u) b_u$, using $(d, u/d) = 1$ for squarefree u. □

The two known cases are the two ends of this space: $w = 1$ gives $b = \delta_1$, $\|b\|_1 = 1$ and (Huang–Li's Lemma 1) $B_w \ll e^{-c\sqrt{\log K}}$; while $w = \log$ gives $b = \Lambda$, $\|b\|_1 \asymp N$ — the complete part *is* the binary Goldbach sum — and $B_w \to -\mathfrak{S}(N) \asymp 1$.

1.3 Results

Section 2 records the exact structure of the space: the per-modulus discrepancy is a dilate (Proposition 3), the signs $\mu(k)$ cancel exactly on the $\log k$ branch (Proposition 4), both known weights are the same sum of dilated walls with different weights (Proposition 5), the untruncated object is the Goldbach count itself (Proposition 8), which is therefore a signed sum of nonnegative layers (Proposition 9) over an ordinary Bombieri–Vinogradov modulus (Proposition 10); and the wall cancels out of the count entirely (Proposition 12).

Section 3 proves the no-go: Lemmas 15 and 17 are the two opposing constraints, Theorem 18 is their consequence, and Theorem 21 shows it survives Elliott–Halberstam. Section 4 closes polynomial weights, which lie outside the hypothesis of Theorem 18. Section 5 records the position of the circle method.

1.4 What is not claimed

- **This is a no-go for one method.** Theorem 18 closes divisor switching with Bombieri–Vinogradov as the only input, over the whole weight space of that method. It is not, and does not claim to be, an obstruction to other methods — in particular it says nothing about the spectral large sieve of [4].
- **The genre is classical.** Bombieri's asymptotic sieve [1] shows that sieve weights alone cannot detect primes, however chosen, and the parity obstruction it isolates is the reason one expects a statement of this shape. What is added here is the precise quantitative form for this route (Note 20).
- **No progress toward Goldbach.** Nothing here bounds $C(N)$, and nothing here is a step toward bounding it.
- **The identities of Section 2 are rearrangements.** They are exact and they are finite; none of them is an estimate, and none of them makes any quantity smaller.

2 The design space, exactly

Proposition 3 (the discrepancy is a dilate)**.** *Unconditionally,*

$$A(N;k) \;=\; \mu(k)\,H(N;k), \qquad H(N;k) \;=\; \sum_{\substack{m<N/k\\(m,k)=1}} \Lambda(N-mk)\,\mu(m). \tag{6}$$

Proof. The class $n \equiv N \pmod k$ is $k \mid N-n$, so writing $N-n = mk$ turns the sum into $\sum_{m<N/k}\Lambda(N-mk)\mu(mk)$; and $\mu(mk)=\mu(m)\mu(k)$ when $(m,k)=1$, zero otherwise. □

Identity (6) qualifies the shape of the whole problem. EH_μ is consumed where one Möbius argument *divides* the other, but the per-modulus discrepancy is $\mu(k)$ times a sum in which the two arguments are related by a *difference*, at scale N/k. The two couplings are not two fields; they are the same field seen before and after the switch. It also explains why μ is the noisier of the two inputs in progressions: for μ, $|A(N;k)| = |H(N;k)|$ is a Möbius–prime correlation of length N/k and nothing makes it small, whereas for a random sign pattern ε the sum $\sum_m \Lambda(N-mk)\varepsilon(mk)$ is a sum of independent signs and exhibits square-root cancellation.

Proposition 4 (the weights are nonnegative)**.** *Substituting* (6) *into* $T_{\log}$ *and using* $\mu(k)^2=1$ *on squarefree* k*,*

$$T_{\log} \;=\; \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)\,(\log k)\,H(N;k) \;-\; C(N)\,B_{\log}(K), \qquad B_{\log}(K) = \sum_{\substack{k<K\\(k,N)=1}} \frac{\mu(k)\log k}{\varphi(k)}. \tag{7}$$

Proof. Immediate from (6) and $\mu(k)\mu(k) = \mu^2(k)$. $\square$

The signs $\mu(k)$ cancel exactly: they cancel against the $\mu(k)$ that (6) extracts from the progression sum, and what is left is a sum of dilated Möbius–prime correlations with *nonnegative* weights $\mu^2(k)\log k$. Huang–Li discard the $\mu(k)$ by the triangle inequality; on the $w_k = 1$ branch keeping them is what makes the unconditional theorem of [6] possible; on the $\log k$ branch they are not there to keep. Whatever smallness $T_{\log}$ has must therefore come from the signs of $H(N;k)$ across k.

Proposition 5 (both ends are the same sum of dilated walls)**.** *For any weight w that is a function of k alone,*

$$T_w \;=\; \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)\, w_k\, H(N;k) \;-\; B_w\, C(N). \tag{8}$$

In particular

$$T_1 = \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)H(N;k) - B_1C(N), \qquad T_{\log} = \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)(\log k)H(N;k) - B_{\log}C(N),$$

and both weights are nonnegative on the range.

Proof. Put (6) into (2). The transform carries $\mu(k)$, so terms with $\mu(k) = 0$ contribute nothing on the left; on the remaining k, $\mu(k)A(N;k) = \mu(k)^2H(N;k)$. The restriction is therefore carried on the right, and the subtracted mean term assembles into $B_wC(N)$ by (4). $\square$

Note 6 (the restriction is not decoration)*.* Dropping $\mu^2(k)$ from (8) does not weaken it, it falsifies it: $H(N;k)$ does not vanish at squarefull k, while the left-hand side does. The terms so admitted are of the size of T_w itself — at $\theta' = 0.56$ they add $5.4 \cdot 10^{-2}N$ at $w = 1$ and $2.9 \cdot 10^{-1}N$ at $w = \log$, the latter being two thirds of $|T_{\log}|$.

Since $B_1 \ll e^{-c\sqrt{\log K}}$ kills the second term while $B_{\log} \to -\mathfrak{S}(N)$ does not, the unconditional theorem of [6] says exactly that the *flat* sum of dilated walls is $\ll_A N(\log N)^{-A}$, and the Goldbach problem says the same sum weighted by $\log k$ is not small. Everything between what is proved and what is open is the factor $\log k$ inside a positively weighted sum of the same terms.

Measurement 7 (the two ends, measured)**.** Over $N = 2 \cdot 10^5$ to $2.56 \cdot 10^7$ by doubling, at $\theta' = 0.56$, (8) holds to a worst relative error of $1.875 \cdot 10^{-16}$, and

$$\frac{\left|\sum_k \mu^2(k)H(N;k)\right|}{\left|\sum_k \mu^2(k)(\log k)H(N;k)\right|} \;=\; 0.1785,\ 0.1559,\ 0.1484,\ 0.1483,\ 0.1367,\ 0.1334,\ 0.1241,\ 0.1135,$$

falling on average; at the top $|T_1|/N = 0.01425$ against $|T_{\log}|/N = 0.1245$. Fitted against $\log N$ the flat sum decays as $N^{-0.3505}$ and $|T_{\log}|/N$ as $N^{-0.2658}$: the gap widens, as it must if one side is $\ll_A N(\log N)^{-A}$ and the other is $\asymp N$. The index set here runs from $k = 1$, as (2) and (8) require; note that $H(N;1) = C(N)$ exactly, so dropping that one term removes $C(N)$ from the flat sum — which changes the ratios above by up to 10.4% and the fitted exponent to -0.3620, while leaving $|T_1|$ untouched to 10^{-18}, since the $k = 1$ term of T_1 is $A(N;1) - C(N)/\varphi(1) = 0$. The two columns are computed side by side, from a sieve built independently of the one `lab_weight_gap.py` uses, in `c02_flatsum_k1.py`; that script's loop is the $k \geq 1$ one, and `lab_weight_gap.py`'s starts at $k = 2$. Identity (6) itself is verified over every squarefree $k < N^{0.56}$ coprime to N at $N = 2 \cdot 10^5$, $4 \cdot 10^5$, $8 \cdot 10^5$ with worst relative error $2.294 \cdot 10^{-14}$, $1.197 \cdot 10^{-13}$, $4.591 \cdot 10^{-14}$, and (7) against a direct evaluation of $T_{\log}$ to at most $1.8 \cdot 10^{-16}$ over $N = 2 \cdot 10^5$ to $3.2 \cdot 10^6$ (`lab_weight_gap.py`, `lab_dilate_identity.py`, `lab_positive_weights.py`).

Proposition 8 (the untruncated sum is the count)**.** *Summing over* all k*, with no truncation and no coprimality restriction,*

$$\sum_{k\geq 1}(\log k)\,\mu(k)\,A(N;k) \;=\; -\sum_{n<N}\Lambda(n)\,\mu(N-n)\Lambda(N-n) \;=\; \sum_{p<N}\Lambda(N-p)\log p. \tag{9}$$

So the untruncated object is $\tilde{r}(N)$*, up to its prime-power part.*

Proof. Both exchanges are finite sums. Since $A(N;k)$ sums n over $k \mid N-n$, exchanging the order and applying $\sum_{d|u}\mu(d)\log d = -\Lambda(u)$ — Möbius inversion of $\log = \mathbf{1} * \Lambda$ — gives the first equality. The second holds because $\mu(u)\Lambda(u) = -\log p$ at $u = p$ prime and vanishes at every higher prime power. □

Proposition 8 is not a decoration: it says that (8) with $w = \log$ is a *rearrangement* of $\tilde{r}(N)$ rather than an approximation waiting for an estimate, so all of its content is in the truncation at K.

Proposition 9 (the count as signed layers)**.** *Cutting the same double sum along the inner variable of* (6)*,*

$$\sum_{k}\mu^2(k)(\log k)H(N;k) \;=\; \sum_{m}\mu(m)\,L(N;m), \qquad L(N;m) := \sum_{\substack{k<N/m,\ (k,m)=1\\ \mu(k)\neq 0}}(\log k)\,\Lambda(N-mk), \tag{10}$$

and every $L(N;m)$ *is nonnegative. With Proposition 8, the Goldbach count is therefore an alternating sum, signed by* μ*, of nonnegative layers.*

Proof. A finite exchange; the coprimality conditions on the pair (k,m) are symmetric. Nonnegativity is that of $\log k$ on $k \geq 1$ and of Λ. □

Proposition 10 (the count over a combined modulus)**.** *Expanding the squarefree condition inside* (10) *by* $\mu^2(k) = \sum_{d^2|k}\mu(d)$ *and writing* $k = d^2 j$*,*

$$\sum_{p<N}\Lambda(N-p)\log p \;=\; \sum_{\substack{m,d\ \text{squarefree}\\ (d,m)=1,\ md^2<N}}\mu(m)\mu(d)\,L_2(N;m,d), \tag{11}$$

where

$$L_2(N;m,d) = \sum_{\substack{j<N/(md^2)\\ (j,m)=1}}\log(d^2 j)\,\Lambda\big(N - md^2 j\big),$$

and L_2 *carries no squarefree condition: it is a prime count in the progression* $p \equiv N \pmod{md^2}$*, an unadorned Bombieri–Vinogradov object of modulus* $q = md^2$*.*

Proof. Finite exchange, and $(k,m) = 1$ splits as $(d,m) = (j,m) = 1$. □

Measurement 11 (how large that modulus has to be)**.** Being an ordinary Bombieri–Vinogradov object is a statement about shape, not about level, and the level is the whole question. Verified over $N = 2\cdot 10^5$ to $1.6\cdot 10^6$ — 166384 to 1331048 pairs — (11) holds to a worst relative error of $4.525\cdot 10^{-15}$. Truncating it to $q < Q$ and letting $Q^*(N)$ be the least Q past which the deficit stays under $0.01N$,

$$Q^* = 160112,\ 200848,\ 482589,\ 589203, \qquad Q^*/\sqrt{N} = 358.0213,\ 317.5686,\ 539.5509,\ 465.8059,$$

so Q^* exceeds $\sqrt{N}$ by two orders at every N tested. Expanding the squarefree condition buys the ordinary modulus and costs one to two orders in its size — which is the barrier again, in the one place the identity had made it measurable (`lab_combined_modulus.py`).

Proposition 12 (the wall cancels out of the count)**.** *Substituting* (7) *into the identity of [6, Thm. 3], the two occurrences of $C(N)$ carry opposite signs and cancel:*

$$\tilde{r}(N) \;=\; \mathfrak{S}(N)N \;+\; \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)(\log k)H(N;k) \;-\; C(N)\big(B_{\log}(K)+\mathfrak{S}(N)\big) \;+\; O_A\bigg(\frac{N}{(\log N)^A}\bigg). \tag{12}$$

Since $B_{\log}(K) \to -\mathfrak{S}(N)$, the quantity $C(N)$ enters the Goldbach count only through the vanishing factor $B_{\log}(K)+\mathfrak{S}(N)$. Consequently

$$\sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)(\log k)\big|H(N;k)\big| \;\le\; (1-\varepsilon)\,\mathfrak{S}(N)\,N \tag{13}$$

suffices for $\tilde{r}(N) > 0$, for all large even N.

Proof. Immediate from (12) and $|C(N)| \le \mathfrak{A}(N)N(1+o(1))$, the last bound now multiplied by $B_{\log}(K)+\mathfrak{S}(N) = o(1)$ rather than by $\mathfrak{S}(N)$. □

Note 13 (what the cancellation buys, and what it does not). Condition (13) has the same shape as the constant-factor condition of [6, Prop. 6]: a bound at level $N^{\theta'}$, with no saving in $\log N$. Nothing has moved off the level axis. What has changed is the threshold constant, from $\mathfrak{S}(N)(1-\mathfrak{A}(N))$ to $\mathfrak{S}(N)$, and the change removes the arithmetic sensitivity along with the slack, because the two were the same thing: $1-\mathfrak{A}(N)$ collapses at primorials, and $\mathfrak{S}(N)$ does not. This is offered as a weaker sufficient condition, proved from two identities. It is not offered as something computation confirms: at accessible N the unspecified O_A term of (12) is numerically the very quantity (13) must bound.

Measurement 14 (the two thresholds, compared)**.** At $\theta' = 0.56$ over $N = 2\cdot 10^5$ to $3.2\cdot 10^6$, with $B_H(N) = \sum_{k<K}\mu^2(k)(\log k)|H(N;k)|$, the new ratio $B_H(N)/(\mathfrak{S}(N)N)$ reads 0.4578, 0.4064, 0.4079, 0.3769, 0.3338 — under 1 throughout and falling — against 2.1519, 1.9105, 1.9175, 1.7720, 1.5691 for the same numerator against the old threshold, $B_H(N)/(\mathfrak{S}(N)(1-\mathfrak{A}(N))N)$, which is above 1 throughout. *The numerator is held fixed across the two lists*; only the threshold constant moves, which is the whole of the comparison. (For the numerator of [6, eq. (9)], which keeps the subtracted mean inside the absolute value and so is a different sum, the old ratio reads 2.1591, 1.9747, 1.9500, 1.7483, 1.5798 — the same conclusion, a different quantity.) Across a seven-point family of arithmetically diverse N the new ratio spans a factor 2.00 where the old spans 23.25. The residual coupling is negligible at these N: $|B_{\log}+\mathfrak{S}|/\mathfrak{S}$ stays under 0.0383 and $|C(N)(B_{\log}+\mathfrak{S})|/N$ never exceeds $3.315\cdot 10^{-4}$, four orders below $\mathfrak{S}(N)$. The residual of (12) decays as $N^{-0.2794}$ with correlation 0.99930 and falls to 2% of N only near $N = 10^{10.16}$ (`lab_direct_route.py`, `lab_direct_identity.py`; the two lists held to one numerator, and the third quantity in the parenthesis, are recomputed side by side in `a1_bh_vs_b.py` — `lab_direct_route.py` prints the parenthetical list, not the two above it).

3 The no-go

3.1 The two opposing constraints

Lemma 15 (extraction)**.**

$$B_w = \sum_{\substack{d<K\\(d,N)=1}} b_d\,\frac{\mu(d)}{\varphi(d)}\,\rho_{dN}\bigg(\frac{K}{d}\bigg), \qquad \rho_n(x) = \sum_{j<x,\,(j,n)=1} \frac{\mu(j)}{\varphi(j)},$$

and $\rho_n(x) \ll e^{-c_1\sqrt{\log x}}$ whenever $\log n \ll \log x$.

Proof. Write $k = dj$; for squarefree k one has $(d, j) = 1$, $\mu(k) = \mu(d)\mu(j)$ and $\varphi(k) = \varphi(d)\varphi(j)$. The bound on ρ is Huang–Li's Lemma 1, a form of Goldston–Yıldırım [2], and the hypothesis $\log n \ll \log x$ is theirs. $\square$

Note 16 (the hypothesis is load-bearing, and it is met where it is used). Without $\log n \ll \log x$ the bound is false: at $\lfloor K/d_0 \rfloor = 1$ Measurement 22 prints $|B_w|\varphi(d_0) = 1.000000$, no saving at all, and that is exactly the corner where $\log n \asymp \log N$ while $\log x = O(1)$. Theorem 18 is unaffected: it maximises only over $d \le N^{\theta'-1/2-\delta}$, where $x = K/d \ge N^{1/2+\delta}$ and $n = dN \le N^{3/2}$, so $\log n \asymp \log x$ holds throughout the range it uses. The same applies to Theorem 21.

Lemma 17 (BV-accessibility)**.** *Expanding* $w_k = \sum_{d|k} b_d$ *inside the residual gives* $\mathcal{R}_w = \sum_d b_d \, \mathcal{R}^{(d)}$*, where* $\mathcal{R}^{(d)}$ *is a sum of* Λ *over progressions to moduli* md *with* $m < N^{1-\theta'}$*. Bombieri–Vinogradov bounds each* $\mathcal{R}^{(d)} \ll_A N(\log N)^{-A}$ *provided* $d \le N^{1/2-\delta}/N^{1-\theta'} = N^{\theta'-1/2-\delta}$*; hence, on that support,* $\mathcal{R}_w \ll_A \|b\|_1 \, N(\log N)^{-A}$*.*

Proof. The residual of the switch is [6, eq. (16)], which after unfolding μ^2 and the coprimality condition, every term is a sum of Λ over a progression to modulus $q = m \operatorname{lcm}(d^2, e)$ with $m < N^{1-\theta'}$ and d, e at most a fixed power of $\log N$; Bombieri–Vinogradov is applied in the form of [5]. Inserting the extra divisor $d \mid k$ multiplies the modulus by d, and the level condition $q \le N^{1/2-\delta}$ becomes the stated bound on d. Summing the $\ll_A N(\log N)^{-A}$ over the support of b with the trivial weight $|b_d|$ gives the claim. $\square$

The two lemmas pull in opposite directions. Lemma 15 says B_w is controlled by the part of b sitting *at* the truncation point K: mass at $d \le K^{1-\varepsilon}$ is damped by $e^{-c\sqrt{\varepsilon \log K}}$. Lemma 17 says b may only live *below* $N^{\theta'-1/2}$. The two thresholds are separated by a factor $N^{1/2}$.

3.2 The theorem

Theorem 18 (no weight extracts $C(N)$)**.** *Fix* $\theta' \in (1/2, 1)$ *and* $\delta > 0$*, and let* w *be any weight whose transform* $b = \mu * w$ *is supported in* $[1, N^{\theta'-1/2-\delta}]$ *— the range in which the residual is accessible to Bombieri–Vinogradov. Then*

$$\frac{\|b\|_1}{|B_w|} \gg \exp\left(c_1 \sqrt{(\tfrac{1}{2} + \delta) \log N} \right). \tag{14}$$

Consequently, even granted $T_w \ll_A N(\log N)^{-A}$ *for every* $A > 0$ *— which is what [6, Thm. 1] supplies unconditionally at* $w = 1$*, and the most that* EH_μ *would supply for a general* w *— identity* (5) *yields at best*

$$|C(N)| \ll_A \exp\left(c_1 \sqrt{\tfrac{1}{2} \log N} \right) \cdot \frac{N}{(\log N)^A},$$

which is not a saving of any power of $\log N$*. No weight in the design space extracts* $C(N)$ *by divisor switching, not even when the hard input is given for free.*

Proof. By Lemma 15 and $\varphi(d) \ge 1$,

$$|B_w| \le \sum_{d \le D} \frac{|b_d|}{\varphi(d)} \left| \rho_{dN}\left(\frac{K}{d}\right) \right| \le \|b\|_1 \max_{d \le D} \left| \rho_{dN}\left(\frac{K}{d}\right) \right| \ll \|b\|_1 \, e^{-c_1\sqrt{\log(K/D)}},$$

with $D = N^{\theta'-1/2-\delta}$, so that $K/D = N^{1/2+\delta}$. This is (14).

For the second assertion, solve (5) for $C(N)$ and bound the three terms on the right. The complete part is $\ll \|b\|_1 \log N$ by Lemma 2 and $\Lambda \ll \log N$; the residual is $\ll_A \|b\|_1 N(\log N)^{-A}$

by Lemma 17, the support hypothesis on b being exactly what that lemma needs; and $T_w \ll_A N(\log N)^{-A}$ is the granted hypothesis. Since $\|b\|_1 \geq |b_1|$ and the first two dominate,

$$|C(N)| \;\leq\; \frac{\|b\|_1}{|B_w|} \cdot O_A\!\left(\frac{N}{(\log N)^A}\right) \;\ll_A\; \exp\!\left(c_1\sqrt{\tfrac{1}{2}\log N}\right)\frac{N}{(\log N)^A},$$

by (14). Finally $e^{c_1\sqrt{\log N/2}}$ exceeds every power of $\log N$, so the saving is destroyed for every A. □

Note 19 (why granting T_w is the right form)*.* Without the granted hypothesis the theorem would be vacuous rather than strong: (5) could then be satisfied by T_w alone being large, and it says nothing about $C(N)$ at all. Granting it isolates what the design space actually fails at. The failure is not a shortage of input — the input is handed over — but the arithmetic of two thresholds that lie a factor $N^{1/2}$ apart. That is why raising the level does not help either (Theorem 21).

The loss factor is $\exp(c_1\sqrt{(1/2)\log N})$, and the $1/2$ in the exponent is literally the exponent of the $\sqrt{N}$ barrier: it is the gap between where a weight must live to see $C(N)$ (at $N^{\theta'}$, the truncation point) and where it may live for Bombieri–Vinogradov to close its residual (below $N^{\theta'-1/2}$). The two known weights exhibit the two endpoints of this space; Theorem 18 shows the interior is empty, and shows why.

Note 20 (relation to Bombieri's asymptotic sieve)*.* The genre of Theorem 18 is classical. Bombieri's asymptotic sieve [1] shows that sieve weights alone cannot detect primes, however the weights are chosen, and the parity obstruction it isolates is the reason one expects a statement of this shape. What Theorem 18 adds is not the phenomenon but its precise form here: the obstruction is quantified as a separation of $N^{1/2}$ between two thresholds, it is specific to the divisor-switch route, and it survives granting the full Elliott–Halberstam conjecture rather than merely Bombieri–Vinogradov. The reason the statement is worth making is narrow but real: without it, the $\log k$ weight looks like one member of a family of candidate consumptions, and the identity of [6, Thm. 3] then looks like an accident of one choice of weight rather than a property of the route.

3.3 The no-go survives Elliott–Halberstam

The obvious objection to Theorem 18 is that Bombieri–Vinogradov is not the only available input: Huang–Li's Theorem 1 itself *assumes EH* for Λ at level N^{θ}. Raising the assumed level does not help.

Theorem 21. *Suppose Λ has level of distribution $\theta_E \in (0,1)$, i.e. $EH(N^{\theta_E})$ holds. Then the residual is accessible only for b supported on $d \leq N^{\theta_E-(1-\theta')}$, while extraction still requires mass at $d \asymp K = N^{\theta'}$; the two thresholds are separated by $N^{1-\theta_E}$, and*

$$\frac{\|b\|_1}{|B_w|} \;\gg\; \exp\!\left(c_1\sqrt{(1-\theta_E)\log N}\right),$$

which exceeds every power of $\log N$ for each fixed $\theta_E < 1$.

Proof. Identical to Theorem 18 with $N^{1/2-\delta}$ replaced by N^{θ_E}: the residual's moduli are md with $m < N^{1-\theta'}$, so $d \leq N^{\theta_E}/N^{1-\theta'}$, whence $K/D = N^{\theta'}/N^{\theta_E-1+\theta'} = N^{1-\theta_E}$, and Lemma 15 gives the ratio. □

Closing the gap would require $\theta_E = 1$ exactly: equidistribution of Λ in progressions to moduli of size N itself, where each progression contains $O(1)$ terms and the statement carries no information. So the route stays closed *even granting the full Elliott–Halberstam conjecture* — it needs not a stronger level but a different mechanism.

Measurement 22 (the load-bearing lemma, checked as an identity)**.** Lemma 15 carries the theorem, so it is checked directly at $N = 99\,999\,998$, $\theta' = 0.56$, $K = \lfloor N^{\theta'} \rfloor = 30199$. For the single-divisor weights $w_k = [\,d_0 \mid k\,]$, i.e. $b = \delta_{d_0}$, the lemma is an identity and is verified as one: over all $10\,262$ squarefree $d_0 < K$ coprime to N, the brute-force B_w over $k < K$ and the factorised $\mu(d_0)\varphi(d_0)^{-1}\rho_{d_0 N}(K/d_0)$ agree to $6.5 \cdot 10^{-18}$.

The size behaves as the lemma predicts. The quantity $|B_w|\varphi(d_0)$ is of order 1 only when $K/d_0 = O(1)$, i.e. only when the weight's transform sits at the truncation point, and is damped throughout the range Bombieri–Vinogradov admits: for these parameters $N^{\theta'-1/2} = 3$, and over $d_0 \le 3$ the largest value attained is 0.004145. The value is not a function of K/d_0 alone — $\rho_{d_0 N}$ carries the condition $(j, d_0 N) = 1$, so it depends on which small primes divide d_0. At $\lfloor K/d_0 \rfloor = 7$ the 184 admissible d_0 produce exactly four values, split by $\gcd(d_0, 15)$:

$\gcd(d_0, 15)$	1	3	5	15
$\lvert B_w\rvert\,\varphi(d_0)$	0.250000	0.750000	0.500000	1.000000

and at $\lfloor K/d_0 \rfloor = 13$ the 55 admissible d_0 produce six values spanning 0.066667 to 0.566667 (`audit_extraction_tradeoff.py`).

4 Smooth weights: $\mu * \log^D = \Lambda_D$

Theorem 18 closes the weights whose transform $b = \mu * w$ is supported low enough for Bombieri–Vinogradov. That hypothesis excludes the most natural family of all, $w_k = f(\log k)$ with f a polynomial, whose transform is spread over every scale. For those the structure is explicit:

$$b \;=\; \mu * \log^D \;=\; \Lambda_D,$$

the generalised von Mangoldt function. Λ_D vanishes on integers with more than D prime factors, and on a squarefree $u = p_1 \cdots p_r$ with $r \le D$ it is the r-th finite difference of x^D at the points $\log p_i$; in particular $\Lambda_D(u) = D!\,\log p_1 \cdots \log p_D$ when $r = D$. Hence for $f = x^D$ the complete part splits by the number of prime factors,

$$\mathrm{CP}_D(N) = \sum_{r=1}^{D} \sum_{\substack{u<N \text{ squarefree} \\ \omega(u)=r}} \Lambda(N-u)\,\Lambda_D(u), \tag{15}$$

the $r = 1$ piece being a Goldbach-type sum and the $r \ge 2$ pieces Chen-type sums (prime plus r-almost-prime).

Proposition 23 (polynomial weights)**.** *Let $w_k = f(\log k)$ with $f = \sum_{j \le D} c_j x^j$ real.*

(i) If $D = 0$ then $B_w \ll e^{-c\sqrt{\log K}}$ (Huang–Li's Lemma 1): no extraction.

(ii) If f is a single monomial x^D, $D \ge 1$, then every term of CP_D is nonnegative, since $\Lambda \ge 0$ and $\Lambda_D \ge 0$; so $\mathrm{CP}_D \ge 0$ and no cancellation is available inside it. *Classical sieve upper bounds give $\mathrm{CP}_D \ll N(\log N)^{D-1}$. A matching lower bound is not available: at $D = 1$ the sum is exactly the binary Goldbach sum $\sum_{p<N} \Lambda(N-p)\log p$, whose order is the assertion to be proved. So CP_D is not known to be $o(N)$, and at $D = 1$ deciding whether it is would settle Goldbach.*

(iii) Cancellation therefore requires coefficients of opposite sign, i.e. tuning c_1 against $c_2, \ldots$; but the ratio that would have to be matched involves the asymptotics of the $r = 1$ piece, which for $D = 1$ is *the binary Goldbach sum. The tuning is circular.*

Consequently no polynomial weight makes the complete part a classically known object.

Proof. (i) is Huang–Li's Lemma 1 applied to $w \equiv 1$. For (ii), $\Lambda_D \geq 0$ on the squarefree integers with $\omega \leq D$ because each $\Lambda_D(u)$ is a finite difference of the convex increasing function $x \mapsto x^D$ at $\log p_1 < \cdots < \log p_r$; every summand of (15) is then a product of nonnegative factors, so no cancellation is available and $\mathrm{CP}_D \geq 0$. The upper bound is the classical sieve upper bound for prime-plus-almost-prime sums, applied to each r-piece. No lower bound of the same order is claimed: the $r = 1$ piece is a prime-plus-prime count, on which the sieve gives none — that is the parity obstruction of Note 20 — and at $D = 1$ it is the whole of CP_1. For (iii), a degree-D tuning has D free coefficients and the constraint that must be solved is an identity between the leading asymptotics of the r-pieces; the $r = 1$ piece with $D = 1$ is $\sum_{p<N} \Lambda(N-p)\log p$, whose asymptotic is the assertion to be proved. □

Note 24 (the canonical tuning moves the wrong way)*.* Since $b = \mu * w$ we have $w = \mathbf{1} * b$ and hence $B(s) = W(s)/\zeta(s)$. That series has a *double* pole at $s = 1$ for every degree-2 f, so no such f removes it and b cannot be made mean-zero; the most a degree-2 tuning can do is kill the second-order term. With $f = x^2 + cx$ one has $W = \zeta'' - c\zeta'$, and neither ζ'' nor ζ' carries a $(s-1)^{-1}$ term.

Measurement 25 (the two pieces are the same order)**.** Nonnegativity, and not a separation of scales, is what closes (ii). At $N = 10^6$, $4 \cdot 10^6$, $1.6 \cdot 10^7$ the two pieces of CP_2 are the same order:

N	10^6	$4 \cdot 10^6$	$1.6 \cdot 10^7$
$r = 1:\ \sum_p \Lambda(N-p)\log^2 p \,/\, N$	22.5145	25.0438	27.4579
$r = 2:\ 2\sum_{pq} \Lambda(N-pq)\log p \log q \,/\, N$	17.3648	19.7926	22.2512
ratio r_2/r_1	0.7713	0.7903	0.8104
$\mathrm{CP}_2/(N\log N)$	2.8866	2.9494	2.9967

The ratio drifts toward 1, not toward 0 or ∞. Calibration: the $D = 1$ column reproduces the Goldbach sum $\sum_p \Lambda(N-p)\log p = 1.7565N$, $1.7633N$, $1.7614N$ against $\mathfrak{S}(N) = 1.76043$, which is the same at all three N because N has the same prime support $\{2, 5\}$ in each case (`audit_polyweight.py`).

5 The circle method has zero margin on $C(N)$

Since the switch is closed over its whole design space, it is worth recording the exact position of the other classical mechanism. Write

$$C(N) = \int_0^1 S_\Lambda(\alpha)\, S_\mu(-\alpha)\, e(-N\alpha)\, d\alpha,$$

where

$$S_\Lambda(\alpha) = \sum_{n \leq N} \Lambda(n) e(n\alpha), \qquad S_\mu(\alpha) = \sum_{m \leq N} \mu(m) e(m\alpha).$$

Proposition 26 (zero margin)**.** *Both standard ways of estimating this integral lie at or above the trivial bound* $|C(N)| \leq \psi(N) \sim N$*:*

(i) $\|S_\Lambda\|_2 \|S_\mu\|_2 = \left(\sum_{n \leq N} \Lambda(n)^2\right)^{1/2} \left(\sum_{m \leq N} \mu^2(m)\right)^{1/2} \sim (6/\pi^2)^{1/2} N (\log N)^{1/2}$*, exceeding the trivial bound by a factor* $\asymp (\log N)^{1/2}$*, which* grows*;*

(ii) any bound of the shape $\sup_\alpha |S_\mu(\alpha)| \cdot \|S_\Lambda\|_1$ *is at least* $\|S_\mu\|_2 \|S_\Lambda\|_1 \gg N^{1/2} \cdot N^{1/2} = N$*.*

Proof. (i) is Cauchy–Schwarz followed by $\sum_{n \leq N} \Lambda(n)^2 \sim N \log N$ and $\sum_{m \leq N} \mu^2(m) \sim 6N/\pi^2$. For (ii), the two factors are bounded below for different reasons. Parseval gives $\sup_\alpha |S_\mu| \geq \|S_\mu\|_2 = (6N/\pi^2)^{1/2}$, an identity-level constraint: no improvement in Möbius exponential-sum technology can lower $\sup_\alpha |S_\mu|$ below $(6/\pi^2)^{1/2} N^{1/2}$. For the second factor Parseval runs the *wrong way*: it gives $\|S_\Lambda\|_1 \le \|S_\Lambda\|_2 \ll (N\log N)^{1/2}$, an upper bound, while the trivial lower bound $\|S_\Lambda\|_1 \ge \left|\int_0^1 S_\Lambda(\alpha)e(-n\alpha)d\alpha\right| = \Lambda(n)$ reaches only $\log N$. The bound $\|S_\Lambda\|_1 \gg N^{1/2}$ that (ii) uses is a theorem of Vaughan [7]. $\square$

Davenport's uniform bound $S_\mu(\alpha) \ll_A N(\log N)^{-A}$ is therefore useless here: it is a saving over N, whereas the pairing needs a bound at the scale $N^{1/2}$, which Parseval forbids. This is the parity obstruction in circle-method language — *the binary problem sits exactly at the trivial bound with no margin* — and it is why the method that settles the ternary problem cannot be pushed to the binary one by any sharpening of the Möbius input.

Measurement 27 (the constants)**.** Computed by exact FFT on a $4N$-point grid at $N = 2^{14}, \ldots, 2^{20}$:

N	2^{14}	2^{16}	2^{18}	2^{20}
$\|S_\mu\|_2/\sqrt{N}$	0.7798	0.7797	0.7797	0.7797
$\sup\|S_\mu\|/\sqrt{N}$	3.058	2.742	2.853	2.801
$\|S_\Lambda\|_1/\sqrt{N}$	1.946	2.084	2.219	2.346
(i) bound$/N$	2.297	2.473	2.639	2.795
$N/\big(\sup\|S_\mu\|\,\|S_\Lambda\|_1\big)$	0.168	0.175	0.158	0.152

The first row reproduces $\sqrt{6/\pi^2} = 0.7797$ exactly, confirming the computation; the last row — the margin that route (ii) would need to exceed 1 — sits near 0.16 and falls over the last three abscissae. Route (i) diverges from the trivial bound like $(\log N)^{1/2}$, as predicted. For reference the object itself is small: $C(N)/N = -0.0105, 0.0001, 0.0059, 0.0032$ at these N (`audit_circle_margin.py`).

6 Summary

- The design space of divisor-switch weights is described exactly by seven finite rearrangements (Propositions 3–12): the per-modulus discrepancy is a dilate, the signs $\mu(k)$ cancel on the $\log k$ branch, and the untruncated object is the Goldbach count itself.
- Theorem 18: no weight whose transform is Bombieri–Vinogradov-accessible extracts $C(N)$, and this holds even when the bound on the switch identity's third term T_w — the object the whole problem is about — is granted for free. The loss is $\exp(c_1\sqrt{(1/2)\log N})$, and the 1/2 is the exponent of the square-root barrier.
- Theorem 21: the separation survives $EH(N^{\theta_E})$ for every fixed $\theta_E < 1$. Closing it would require $\theta_E = 1$, where the statement is vacuous.
- Proposition 23: polynomial weights, which lie outside the hypothesis of Theorem 18, are closed by nonnegativity of Λ_D and by circularity of the tuning.
- Proposition 26: the circle method applied directly to $C(N)$ has zero margin, with Parseval supplying the floor.
- Net progress toward Goldbach: zero. What is closed is a design space, so that it need not be re-explored.

References

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[3] J.-J. Huang and H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811 [math.NT]; v1 (May 2020), v2 (August 2022). Also published in a Springer volume, `doi:10.1007/978-3-030-67996-5_17`. References here are to the arXiv version, which we have read, and not to the published version, which we have not.

[4] J. D. Lichtman (with an appendix by S. Drappeau), *Primes in arithmetic progressions to large moduli, and Goldbach beyond the square-root barrier*, arXiv:2309.08522 [math.NT], 2023.

[5] H. Montgomery and R. Vaughan, *Multiplicative number theory I: classical theory*, Cambridge University Press, 2007.

[6] *An unconditional bound for a Möbius-weighted correlation sum in fixed residue classes, with two consequences for the Huang–Li reduction of Goldbach to Elliott–Halberstam*, companion paper.

[7] R. C. Vaughan, *The L^1 mean of exponential sums over primes*, Bull. London Math. Soc. **20** (1988), 121–123.

A Reproducibility

Script	Result file	Supports
`lab_dilate_identity.py`	`lab_dilate_identity.txt`	Proposition 3
`lab_positive_weights.py`	`lab_positive_weights.txt`	Proposition 4
`lab_weight_gap.py`	`lab_weight_gap.txt`	Measurement 7, $k \geq 2$ column
`c02_flatsum_k1.py`	`c02_flatsum_k1.txt`	Measurement 7, $k \geq 1$ column
`lab_direct_identity.py`	`lab_direct_identity.txt`	Proposition 8
`lab_layer_decomposition.py`	`lab_layer_decomposition.txt`	Proposition 9
`lab_combined_modulus.py`	`lab_combined_modulus.txt`	Proposition 10
`lab_direct_route.py`	`lab_direct_route.txt`	Measurement 14, residual and paren
`a1_bh_vs_b.py`	`a1_bh_vs_b.txt`	Measurement 14, the two lists
`audit_extraction_tradeoff.py`	`audit_extraction_tradeoff.txt`	Measurement 22
`audit_polyweight.py`	`audit_polyweight.txt`	Measurement 25
`audit_circle_margin.py`	`audit_circle_margin.txt`	Measurement 27

No computation is used as a premise of any proof. The identities of Section 2 are finite rearrangements and are verified as such to machine precision; the measurements record sizes over the ranges stated in each statement.